

IRIS - Full Parts List and Costs

Everything in the robot: the moving body, the S3 face-and-senses board, the ESP32-CAM eye, wiring, tools, and every piece of software with its licence and price. Prices are in Indian rupees.

Cost summary

Section	Required	Optional extras
1. The brain - your laptop	Rs 0	-
2. Robot body - movement (firmware: esp32-iris-node-bts7960)	Rs 5,890	-
3. Face and senses - ESP32-S3 board (firmware: esp32-s3-iris-sensors)	Rs 2,625	-
4. The eye - ESP32-CAM (firmware in the cammodule repo)	Rs 720	Rs 250
5. Voice on the robot itself - optional	Rs 0	Rs 580
6. Wiring, mounting and consumables	Rs 890	Rs 300
7. Tools (skip if you already own them)	Rs 0	Rs 1,270
Software, hosting, AI services (section 8)	Rs 0	Rs 450 one-off
Total	Rs 10,125	Rs 2,850
Everything, required + all optional	Rs 12,975	

Bottom line: about **Rs 10,125** buys every required part for the complete robot with movement, eyes, all sensors and the camera. The software costs nothing: IRIS runs on your own laptop, the AI models run on free tiers, and there is no server to rent. Recurring cost: **Rs 0 per month**.

Legend: **owned** = you already have it, counted as Rs 0. **(optional)** rows are not in the required total.

1. The brain - your laptop

IRIS itself runs here: the FastAPI server, the dark UI, the microphone, face recognition and the AI gateway. No VPS, no server rental. Any laptop from the last 6-7 years works; 8 GB RAM is comfortable, 16 GB if you want local speech-to-text (faster-whisper) and insightface at the same time.

Part	Details	Where	Qty	Unit	Total
Laptop / PC (already owned)	Windows, macOS or Linux. Python 3.11+. Joins the same phone hotspot as the boards.	Existing	1	owned	Rs 0
Phone with mobile hotspot (already owned)	2.4 GHz hotspot. All three ESP32 boards and the laptop join it; IRIS talks to the boards by IP address.	Existing	1	owned	Rs 0

2. Robot body - movement (firmware: esp32-iris-node-bts7960)

Four-wheel skid steer. One ESP32 drives two BTS7960 half-bridge modules, one per side. IRIS sends one UDP datagram per command (port 8267), which is why it moves with no visible lag.

Part	Details	Where	Qty	Unit	Total
ESP32 DevKit V1 (ESP32-WROOM-32, 30 pin)	Motor controller. WiFi + UDP fast path + calibration web page. Board profile: 'ESP32 Dev Module'.	robu.in / Amazon.in	1	Rs 450	Rs 450
BTS7960 43 A motor driver module (IBT-2)	One per side. VCC needs 5 V; R_EN + L_EN tied to one GPIO each module. Handles the stall current of two gear motors.	robu.in / Robocraze	2	Rs 550	Rs 1,100
12 V DC geared motor, 100-300 RPM, with wheel	Yellow 'BO' motors are too weak for 4WD skid steer on carpet; use 12 V metal-gear or 25GA motors with 65-85 mm wheels.	robu.in	4	Rs 380	Rs 1,520

Part	Details	Where	Qty	Unit	Total
4WD robot chassis plate / frame	Acrylic or aluminium base with motor mounts and a second deck for the S3 board and the head. Pick one sized for your motors.	Amazon.in / robu.in	1	Rs 750	Rs 750
Battery: 3S Li-ion pack 11.1 V 2200-3000 mAh (with BMS)	Motor supply through B+/B- of both BTS7960s. Alternative: 3x 18650 cells + 3S holder + 3S BMS (about the same price).	robu.in	1	Rs 1,300	Rs 1,300
3S Li-ion balance charger 12.6 V 1-2 A	Charges the pack. Do not run the robot from a laptop USB or a power bank on the motor side.	Amazon.in	1	Rs 450	Rs 450
Rocker switch + inline fuse holder + 10 A fuse	Main power cut-off between battery and drivers. The fuse is what saves a BTS7960 when a wire slips.	Local electronics shop	1	Rs 120	Rs 120
XT60 / screw terminal connectors, 18 AWG silicone wire (2 m red + black)	Heavy motor current path: battery -> switch -> fuse -> B+/B-. Signals use normal jumper wire.	robu.in	1	Rs 200	Rs 200

3. Face and senses - ESP32-S3 board (firmware: esp32-s3-iris-sensors)

Two OLED eyes plus every sensor, all on one 3.30 V buck converter so no resistor or divider is needed. Pin map is fixed to your current wiring (ultrasonics on 4-11, DHT22 on 12, PIR 13, flame 14, MQ-2 DO 17, eyes on 20/21 and 38/39).

Part	Details	Where	Qty	Unit	Total
ESP32-S3 DevKitC-1 (N16R8, 44 pin, USB-C)	Eyes + sensors node. Board profile: 'ESP32S3 Dev Module'. The N16R8 (16 MB flash, 8 MB PSRAM) variant is the common one.	robu.in / Amazon.in	1	Rs 950	Rs 950
0.96 inch OLED 128x64, SSD1306, I2C (0x3C)	The eyes. One panel per I2C bus (twin mode) or both on one bus showing the same picture. Blue or white.	robu.in	2	Rs 260	Rs 520
HC-SR04 ultrasonic distance sensor	Two front, two rear. Runs at 3.3 V on genuine modules; some clones need 5 V - buy the '3.3-5 V' version (or RCWL-1601).	robu.in	4	Rs 85	Rs 340
DHT22 / AM2302 temperature + humidity module (white)	On GPIO 12. Read every 2 s by the firmware. DHT11 (blue) works too for about Rs 90 but is less accurate.	robu.in	1	Rs 260	Rs 260
HC-SR501 PIR motion sensor	Room-wide motion on GPIO 13. Set its jumper to 'single trigger' and turn the delay knob fully down.	robu.in	1	Rs 95	Rs 95
Flame / IR fire sensor module (digital + analog out)	DO on GPIO 14. Turn the module's pot until its DO LED goes off with no flame present - otherwise it reports fire at rest.	robu.in	1	Rs 65	Rs 65
MQ-2 gas / smoke sensor module	DO on GPIO 17 (level set by the module knob). Needs a 24-48 h burn-in the first time; runs warm - keep it off the plastic deck.	robu.in	1	Rs 160	Rs 160
LDR light sensor module (photoresistor + comparator)	Room brightness. Wired but currently ignored on GPIO 18 (ADC2 is off under WiFi); move to GPIO 1 to read a percentage.	robu.in	1	Rs 45	Rs 45
LM2596 adjustable buck converter (set to 3.30 V)	Powers the S3, both OLEDs and every sensor from the 12 V pack. Set with a multimeter BEFORE connecting anything.	robu.in	1	Rs 95	Rs 95
Mini-360 / LM2596 buck converter (set to 5.0 V)	5 V rail for the two BTS7960 logic sides and the ESP32-CAM. Same rule: set it first.	robu.in	1	Rs 95	Rs 95

4. The eye - ESP32-CAM (firmware in the cammodule repo)

Face recognition and 'what is this?'. Only two wires: 5 V and GND. It joins the hotspot on its own and IRIS pulls JPEG frames over HTTP; the thinking happens on the laptop.

Part	Details	Where	Qty	Unit	Total
ESP32-CAM (AI-Thinker) with OV2640 2 MP camera	Mount on the head, lens facing forward. Pulls up to 300 mA when transmitting - feed it from the 5 V buck, never the S3's 3.3 V pin.	robu.in / Amazon.in	1	Rs 560	Rs 560
ESP32-CAM-MB programmer / USB shield	Micro-USB board the camera plugs into for flashing (upload at 115200). Often bundled with the camera for about Rs 650 total.	robu.in	1	Rs 160	Rs 160
OV2640 wide-angle (120 deg) lens module - optional swap (optional)	The stock 66 deg lens sees a narrow cone; a wide lens catches you off-centre so 'who am I' works from more of the room.	robu.in	1	Rs 250	Rs 250

5. Voice on the robot itself - optional

IRIS already listens and speaks through the laptop's mic and speakers. These let the robot hear and talk from its own head. Wired to the S3 free I2S pins (38-42, 21 - only in shared-bus eye mode).

Part	Details	Where	Qty	Unit	Total
INMP441 I2S MEMS microphone (optional)	Omnidirectional digital mic; noise-free over I2S. Mounted at the front, away from the motors.	robu.in	1	Rs 260	Rs 260
MAX98357A I2S 3 W class-D amplifier (optional)	Drives a small speaker directly from the S3's I2S output.	robu.in	1	Rs 230	Rs 230
Speaker 3 W 4 ohm (40-50 mm) (optional)	Fits behind a grille on the chest.	robu.in	1	Rs 90	Rs 90

6. Wiring, mounting and consumables

The boring part that decides whether it works on the first day. Jumper wires from a kit are fine for signals; motor current goes through the thick silicone wire in section 2.

Part	Details	Where	Qty	Unit	Total
Dupont jumper wire kit (M-M, M-F, F-F, 20 cm, 120 pcs)	All sensor and OLED signal wiring.	robu.in	1	Rs 180	Rs 180
Breadboard 830 pt + mini breadboards (2)	Common 3.3 V and GND rails for the sensors; the two minis go on the head for the OLEDs.	robu.in	1	Rs 160	Rs 160
USB cables: Micro-USB x2 (ESP32, CAM-MB) + USB-C x1 (S3)	Data cables, not charge-only ones - a charge-only cable is the number one 'board not detected' cause.	Amazon.in	1	Rs 250	Rs 250
M3 nylon standoffs + screws + nuts assortment	Board mounting on the chassis decks.	robu.in	1	Rs 150	Rs 150
Double-sided foam tape, zip ties, heat-shrink tubing	Sensor mounting and strain relief.	Local shop	1	Rs 150	Rs 150
Head / eye enclosure (3D print or acrylic box, 10x8x8 cm) (optional)	Holds the two OLEDs, camera, PIR and the two front ultrasonics. A 3D-print shop charges by weight; about 60-80 g of PLA.	Local 3D print shop	1	Rs 300	Rs 300

7. Tools (skip if you already own them)

Part	Details	Where	Qty	Unit	Total
Soldering iron kit 60 W with stand, solder, flux (optional)	Needed for the OLED headers, battery leads and motor terminals.	Amazon.in	1	Rs 550	Rs 550
Digital multimeter (optional)	Setting the two bucks to 3.30 V and 5.0 V is not optional; guessing has already cost people an S3.	Amazon.in	1	Rs 420	Rs 420
Wire stripper / cutter, small screwdriver set (optional)		Local shop	1	Rs 300	Rs 300

8. Software, services and hosting

Every program, library and online service the robot uses, with its licence and what it costs. All required software is free and open source; the AI runs on providers' free tiers with automatic fallback between them.

Software / service	Licence / plan	Cost	Role and notes
IRIS software (this repository)	Own code	Rs 0	FastAPI backend, dark hologram UI, 83 tools, NLU rules, memory, scheduler, device transport, vision package.
Python 3.11+	PSF licence	Rs 0	Runtime for IRIS.
FastAPI, uvicorn, httpx, pydantic, SQLAlchemy + aiosqlite	MIT / BSD	Rs 0	Web server, device HTTP/UDP transport, settings, persistent memory (SQLite file, no database server).
Arduino IDE 2 + ESP32 board package (Espressif)	GPL / LGPL	Rs 0	Flashing all three boards. Libraries: Adafruit SSD1306 + GFX, DHT sensor library, ArduinoJson - all free.
Face recognition backend: insightface + onnxruntime (recommended) or face-recognition (dlib) or opencv-python-headless (detect only)	MIT / BSD / Apache	Rs 0	Runs on the laptop. Faces never leave the machine; stored in faces.json in the IRIS data folder.
Vision model for 'what is this' - OpenRouter free models (e.g. meta-llama/llama-4-scout:free) or Google AI Studio Gemini Flash free tier	Free tier	Rs 0	Set VISION_MODEL in .env. Free tiers are rate-limited (roughly 50-200 requests/day per provider); IRIS falls over to the next provider automatically.
Conversation LLM - OpenRouter, Groq, Gemini, Cerebras, Mistral, Together, GitHub Models, Hugging Face	Free tiers	Rs 0	Any one key unlocks full conversations; several keys give the circuit-breaker more to fall back on. Offline mock model always works for commands.
Speech-to-text: faster-whisper (local) or browser speech / Google free	MIT	Rs 0	Local STT wants ~2 GB RAM for the 'small' model.
Text-to-speech: edge-tts (online, free) or pyttsx3 (offline)	Free	Rs 0	Natural voices without an account.
Wake word: openwakeword	Apache 2.0	Rs 0	'Hey Iris' detection on the laptop.
Hosting / VPS / domain	Not needed	Rs 0	Everything runs on the laptop over the hotspot. (A VPS would have been about Rs 400-700 per month and cannot use the microphone over plain http anyway.)
Optional: paid API credit if free limits bite	Pay as you go	Rs 450 (about USD 5) one-off	USD 5 of OpenRouter credit lasts months at hobby usage. Only if you exhaust the free tiers during a demo day.
Software total (required)		Rs 0	Optional API credit Rs 450 one-off, only if free limits are hit.

9. Power budget - why two buck converters

Motors: 12 V pack straight into both BTS7960 B+/B- (4 motors, 1-2 A each running, more at stall). 5 V rail (Mini-360): two BTS7960 logic sides (~50 mA) + ESP32-CAM (180 mA, 300 mA peaks) - about 0.5 A. 3.3 V rail (LM2596 at 3.30 V): ESP32-S3 (~250 mA with WiFi) + 2 OLEDs (~40 mA) + 4 HC-SR04 (~60 mA) + PIR, DHT22, flame, LDR (~20 mA)

+ MQ-2 heater (~150 mA) - about 0.6 A. Every GND is common. Nothing on the S3 side ever sees more than 3.3 V, which is why no resistor or voltage divider appears in this list.

10. Notes on the prices

Prices are typical September 2026 Indian retail with GST, without shipping (budget Rs 100-250 per order). Buying the sensors and the ESP32 boards from one seller in a single order usually lands 10-15% under the totals above. The three items that vary most are the motors, the battery pack and the chassis - they depend on the size of robot you build; the figures here are for a 25-30 cm 4WD base that can carry the head and sensor decks. Cheaper substitutions that still work with the firmware: DHT11 for DHT22 (Rs 90), a single OLED (drop Rs 260), 2WD with two motors and one BTS7960 (saves about Rs 1,300).